15 Polly wants to estimate the number of fish in a lake.

On Monday, Polly catches 200 fish.
She puts a tag on each fish and then releases them.

On Tuesday, Polly catches 90 fish.
15 of these fish have a tag on them.

Work out an estimate for the total number of fish in the lake.

Tagged fraction in second catch = 15/90 = 1/6

1 mark
Estimate the same tagged proportion in the lake.

200 tagged fish represent about 1/6 of all fish.

1 mark
Scale the tagged count by 6.

Estimated total = 200 × 6 = 1200

1 mark
Assume tags mix evenly and the population is stable.

1200

(Total for Question 15 is 3 marks)

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